

Heuristic Evaluation and Redesign of the 12306 Home Ticket-Search Flow

A task-first redesign based on Nielsen heuristics: reduce advertising dominance, clarify the primary path, disclose secondary services progressively, and prevent wrong-route or wrong-date mistakes.

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1 Selected System and Task

System: China Railway 12306 mobile app home page.

Purpose: train-ticket search, route comparison, passenger selection, payment, refund, and order management.

Representative task: search a one-way train ticket from Hefei South to Shengzhou Xinchang on June 10.

10

Nielsen heuristics checked

1

home-page task flow inspected

8

prioritized issues found

2 Evaluation Objectives

- Identify where the first screen distracts from the ticket-search goal.
- Separate core booking actions from commercial and secondary services.
- Reduce cognitive load before the user reaches the train-list page.
- Propose a redesign that preserves functions but improves task priority.

3 Methodology

- Walk through the real 12306 home page and map the primary booking path.
- Evaluate the interface against Nielsen's ten usability heuristics.
- Inspect each task step for feedback, consistency, memory demand, error prevention, and recovery.
- Assign severity: cosmetic, minor, major, or catastrophic.

6 System-Level Findings

Strength: the home page exposes ticket search quickly, uses familiar station/date fields, and supports switching between transport modes.

Weakness: the same first screen carries advertising, finance search, service shortcuts, announcements, floating controls, and bottom navigation, which fragments attention before search.

4 Task Flow and Current Interface

Task: search a one-way train ticket from Hefei South to Shengzhou Xinchang on June 10 from the 12306 home page.

1. Land
Open home page with ad and search area
ad dominance

2. Choose
Stay in train-ticket tab
mode choice

3. Enter
Check stations, date, student option
form clarity

4. Query
Tap query and read history
decision focus

5. Continue
Move to train list and booking
flow break



1. Advertising dominance

The red reward banner attracts more attention than the ticket-search task.

2. Fragmented intent

Finance search, pet shipping, utility icons, tabs, and service grid compete on the same screen.

3. Weak contextual help

Student-ticket rules and announcements are visible but not explained at the decision point.

4. Attention residue

Bottom navigation and floating controls remain active while the user is completing the form.

7 Redesigned Home Page Prototype



Task-first hierarchy

The ticket card becomes the strongest visual object; ad height is reduced.

Progressive disclosure

Core travel tools stay visible; commerce services move under "More Services".

Contextual decision support

Student-ticket rules and notices appear exactly where decisions happen.

Error prevention

Route, date, and passenger type persist after search to prevent wrong-route mistakes.

Accessibility

Larger labels, fewer simultaneous options, and a clearer elderly-friendly path.

8 Updated Optimization Plan

Collapse non-task content

Promotion, finance, hotel, shopping, rental, insurance, and tourism become secondary after query.

Protect the main form

Departure, arrival, date, student option, and query button form one clear action cluster.

Route notices by relevance

Only payment, rule, or ticket-validity notices interrupt the search flow.

Measure success

Faster first query, fewer wrong-route/date errors, lower workload, and higher completion rate.

9 Expected UX Impact

The redesign does not remove 12306 functions. It reorganizes attention so the ticket-search task is completed first, while secondary services remain available after the user's main goal is protected.

5 Prioritized Usability Issues

ID	ISSUE	VIOLATED HEURISTIC	SEVERITY
P1	Top advertising dominates the first screen and pushes the booking task downward.	Aesthetic and minimalist design; visibility of status	Major
P2	Financial search, pet shipping, utility icons, and transport tabs weaken the primary path.	Consistency, user control and freedom	Major
P3	Service shortcuts appear before the user completes the ticket-search task.	Flexibility and efficiency; minimalist design	Major
P4	Announcements and promotions interrupt attention but are not connected to the current task.	Visibility of status; help and documentation	Major
P5	History routes are useful but visually weak below the main query button.	Recognition rather than recall	Minor
P6	The student-ticket checkbox lacks in-context eligibility explanation.	Help and documentation	Minor
P7	Many icons use similar visual weight even when their task value differs.	Consistency and standards	Minor
P8	Bottom navigation stays visually active while the main form is active.	Attention management	Cosmetic